

M. Tuna Alatan

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Education

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| Bilkent University , B.S. in Electrical and Electronics Engineering | Sept 2023 – June 2027 |
| • GPA: 3.77/4.0 | |
| METU Foundation School , High School Degree | Sep 2019 – June 2023 |
| • GPA: 93.9/100 | |

Experience

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| Computer Vision Intern , ROMER – METU, Ankara | Jul 2025 – Aug 2025 |
| • Designed a novel pipeline for labeling 3D-LiDAR data by utilizing stereo-vision , LiDAR-Gate, which earned the "Best Presentation of the Internship" award. | |
| • Built an end-to-end perception pipeline using FoundationStereo (depth) , YOLO & SegFormer (segmentation) , DBSCAN (clustering) , and PCA (primitive fitting) . | |
| • The GitHub repository is available at github.com/tuna-alatan/LiDAR-Gate . | |
| Undergraduate Research Intern , METU, Ankara | Jun 2025 – Aug 2025 |
| • Under the supervision of Prof. Dr. Ahmet Oğuz Akyüz conducted research on low-light image enhancement and color calibration using color conversion matrix. | |
| • Conducted a comparative evaluation of low-light image enhancement methods— Histogram Equalization , Plateau Equalization , and CLAHE —focusing on visual quality and computational efficiency. | |
| • Deployed Zero-DCE on Raspberry Pi + Hailo AI Hat for real-time low-light enhancement at the edge. | |
| • Performed color calibration on low-light images captured from low-light-sensitive camera using CCM and ColorChecker . | |
| • The GitHub repository is available at github.com/tuna-alatan/LowLightCalib . | |
| Embedded Systems Intern , Artron – Hacettepe Teknokent, Ankara | Jul 2024 – Aug 2024 |
| • Implemented basic image processing algorithms such as filtering and bilinear interpolation for FPGAs using VHDL . | |
| • Designed image processing pipelines including timing signal generation , test image synthesis , and block RAM-based buffering . | |
| • Simulated image data paths and timing behavior using ModelSim . | |
| • The GitHub repository which contains the code and the internship report can be accessed through github.com/tuna-alatan/Intro-Image-Processing-VHDL . | |
| Computer Vision Intern , OGAM – METU, Ankara | June 2021 – Aug 2021 |
| • Worked with Intel RealSense D435 depth camera using Python SDK for data acquisition and processing. | |

Awards

- **40% Merit Scholarship**, Ihsan Dogramaci Bilkent University, (2025-2026 Academic Year)
- **80% Merit Scholarship**, Ihsan Dogramaci Bilkent University, (2024-2025 Academic Year)
- **90% Merit Scholarship**, METU Foundation High School
- **2nd Place**, RIDERS Robotic League 2022 Season Final

Projects

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| Smart Camera System for Real-Time Wi-Fi Streaming and Target Tracking | GitHub Repository |
| • Developed a smart camera system that is mounted on two-axis servos which can be controlled via a joystick. | |
| • The image data is streamed through Wi-Fi which can be accessed by devices that are connected to the local | |

Wi-Fi.

- A **tracking algorithm** consisting of integral image and **filtering** is embedded on the camera to track a black crosshair in a frame. The location of the crosshair in the frame is then mapped to an 8x8 LED matrix.
- Tools Used: KL25Z ARM Cortex M0+ MCU, ESP32-Cam Module, Embedded C.

Color Tracking Camera

[GitHub Repository](#)

- Developed a **color-tracking** camera system that is controlled by an **FPGA**. A webcam and a servo motor is used to construct the rotating camera hardware. Python and VHDL is used for developing the software.
- Tools Used: BASYS3 FPGA, VHDL, Python.

Ray Tracing Engine

- Implemented a physically based ray tracer with **GGX BRDFs** and experimented with **CUDA** acceleration for performance optimization.
- Tools Used: C++, CUDA.

Technologies

Languages: C, C++, Python, VHDL, 8051 Assembly, CUDA, MATLAB.

Embedded & Hardware: FPGA (Vivado), ModelSim, LTspice, Proteus

Computer Vision & ML: OpenCV, YOLO, SegFormer, PyTorch, Numpy, FoundationStereo.

Creative & Visualization: Blender, Unreal Engine, Substance Painter, Adobe Photoshop, DaVinci Resolve.